Question 1 Solution

The Wilcoxon Rank Sum test is appropriate here since we are dealing with two independent samples of quantitative data, and we are comparing their locations with one another.

H_0: There is no difference in histamine levels between the two groups

H_1: There is a difference in histamine levels between the two groups

<u>Test Statistic:</u>

Ranking:

With allergies:	7	3.5	8			$T_1 = 18.5$
Without:	2	3.5	6	1	5	$T_2 = 17.5$

Since both $n_1 = 3$ and $n_2 = 5 < 10$ we do the Small sample test

Test statistic, $T = T_1 = 18.5$

<u>Critical Region</u>:

From the tables, reject the null hypothesis if $T < T_L = 7$ OR $T > T_U = 20$

<u>Conclusion</u>:

Since $7 < T < 20$, we do NOT Reject the null hypothesis, and can conclude that there is no difference in histamine levels between the two groups

Question 2 Solution

a) The Wilcoxon signed rank sum test is an appropriate non-parametric test because:
   i. We have two paired samples of quantitative data
   ii. Observations within samples are independent

b) H_0: There is no difference between the assessed and market values of the properties
   H_1: The assessed values of the properties are higher than the market values

c) Calculating the test statistic:

Property	Difference	|Difference|	Rank	Signed Rank
1	65	65	2	+2
2	-150	150	5.5	-5.5
3	-285	285	9	-9
4	85	85	3	+3
5	-150	150	5.5	-5.5
6	150	150	5.5	+5.5
7	-200	200	8	-8
8	25	25	1	+1
9	300	300	10	+10
10	-150	150	5.5	-5.5

$T = -12$

n = 10

$$\sigma = \sqrt{\frac{10(11)(21)}{6}}$$

$$\sigma = 19.62$$

$$z = \frac{-12-0}{19.62} = -0.61$$

d) Critical Region: Reject H_0 if $z > 1.96$

e) Since -0.61 < 1.96 we do NOT Reject H_0 and conclude that there is no difference between the assessed and market values of the properties.

<u>Question 3 Solution</u>

The Kruskal-Wallis test is appropriate here since we are comparing three independent populations of quantitative (ratio-scaled) data with respect to their locations.

H_0: The three methods of instruction are equally effective

H_1: The three methods of instruction are not all equally effective

<u>Test statistic:</u>

	Grd	*Comb. Rank*	Grd	*Comb. Rank*	Grd	*Comb. Rank*	Grd	*Comb. Rank*	Grd	*Comb. Rank*	Grd	*Comb. Rank*	Grd	*Comb. Rank*		
A	94	***17***	88	***14***	91	***16***	74	***6***	87	***13***	97	**18**			84	**T1**
B	85	***12***	82	***10***	79	***8***	84	***11***	61	***1***	72	**4.5**	80	**9**	55.5	**T2**
C	89	***15***	67	***2***	72	***4.5***	76	***7***	69	***3***					31.5	**T3**

$$H = \frac{12}{18(19)}\left[\frac{84^2}{6} + \frac{55.5^2}{7} + \frac{31.5^2}{5}\right] - 3(19) = 6.67$$

<u>Critical region:</u>

α = 0.05 (this was not stated in the question – so you choose a reasonable level of significance – 5% is usually a good choice)

df =k−1=3−1=2

From Chi-square tables:

0.025 < p-value < 0.05

<u>Conclusion:</u>

We reject the null hypothesis since the p-value < 0.05, and conclude that the three methods are not all equally effective at the 5% significance level